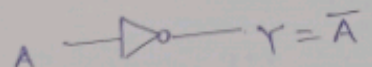
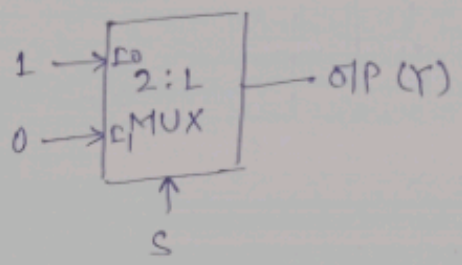
* Implementation of NOT Gate using 2:1 MUX :→



Truth Table :→

I/P A	O/P Y = Ā
0	1
1	0



Truth-Table :→

I/P S	O/P Y
0	1
1	0

$Y = \bar{S} \cdot 1 + S \cdot 0$

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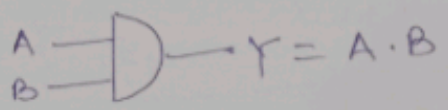
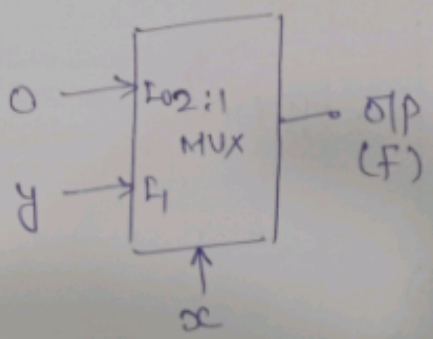
* Implementation of AND Gate using 2:1 MUX :→



Truth Table :→

I/P A	I/P B	O/P Y = AB
0	0	0
0	1	0
1	0	0
1	1	1



Truth-Table :→

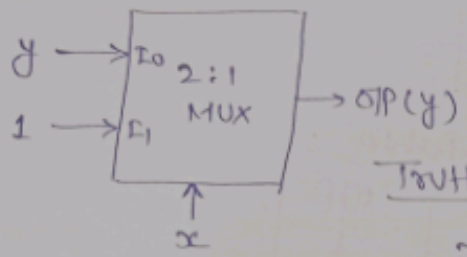
x	y	f	f → y
0	0	0	f = 0
0	1	0	
1	0	0	f = y
1	1	1	

VIVO Y35

* Implementation of OR Gate using 2:1 MUX :→

Truth-Table :→

I/P A	I/P B	O/P Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1



Truth-Table :→

x	y	f	f→y
0	0	0	f=y
0	1	1	
1	0	1	f=1
1	1	1	

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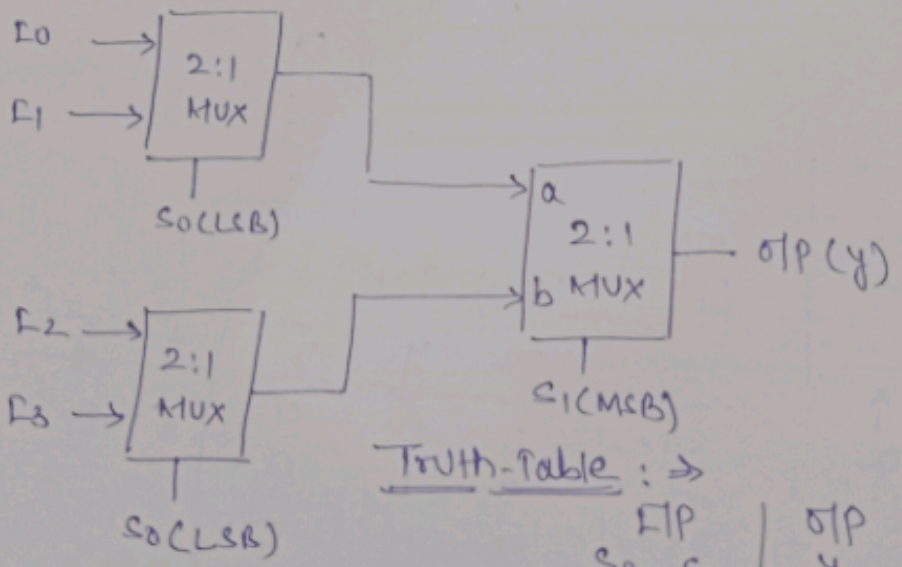
* Implement 4:1 MUX using 2:1 MUX



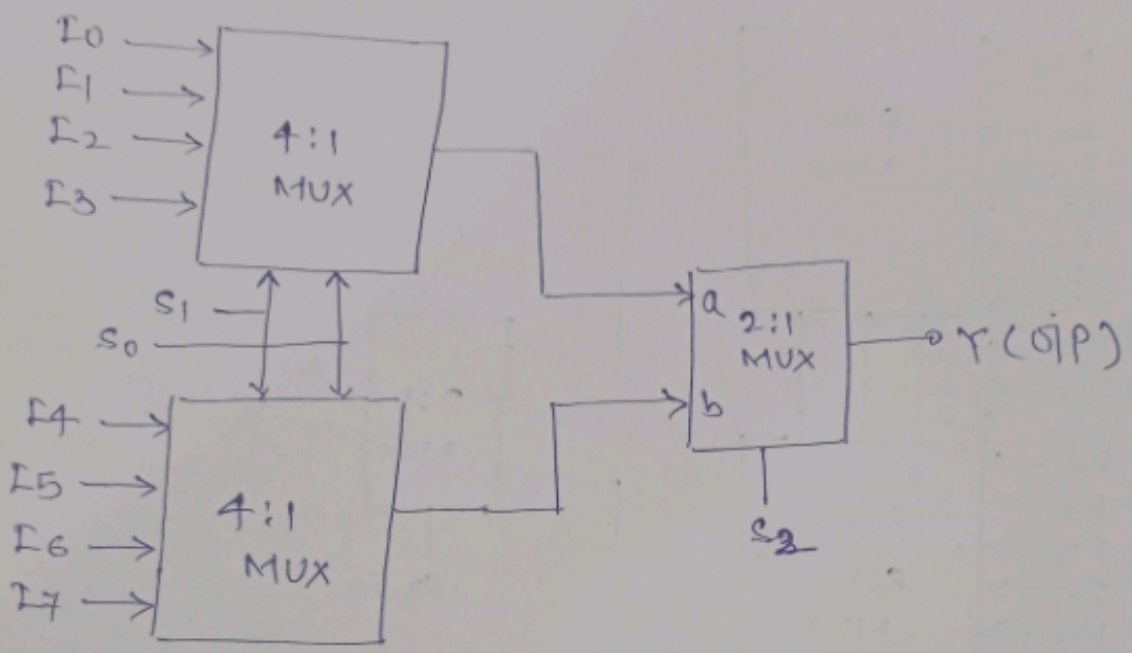
Truth-Table :→

I/P S_0	I/P S_1	O/P y
0	0	I0
0	1	I1
1	0	I2
1	1	I3

VIVO Y35

... using 2:1 MUX :→ ①

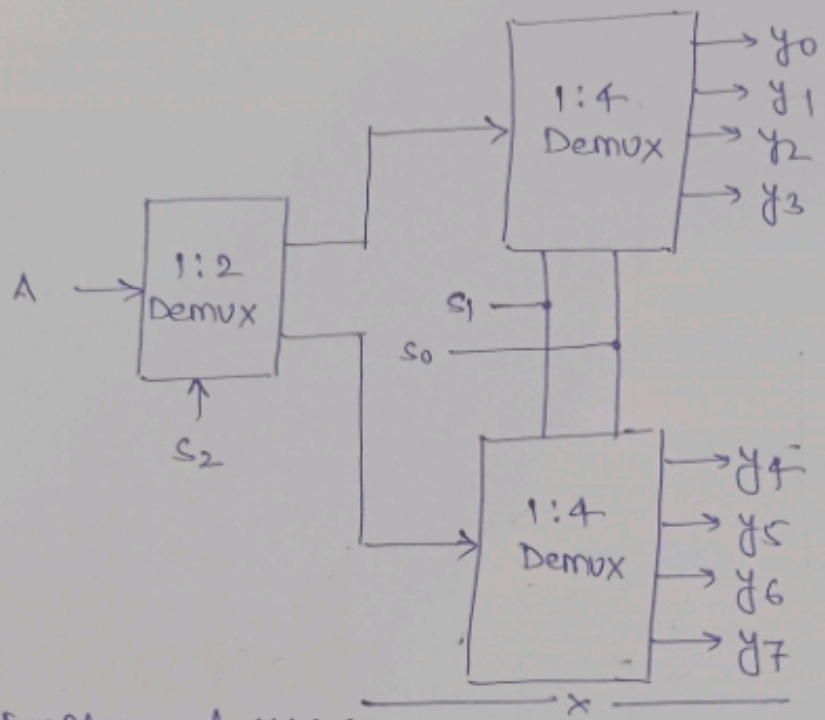
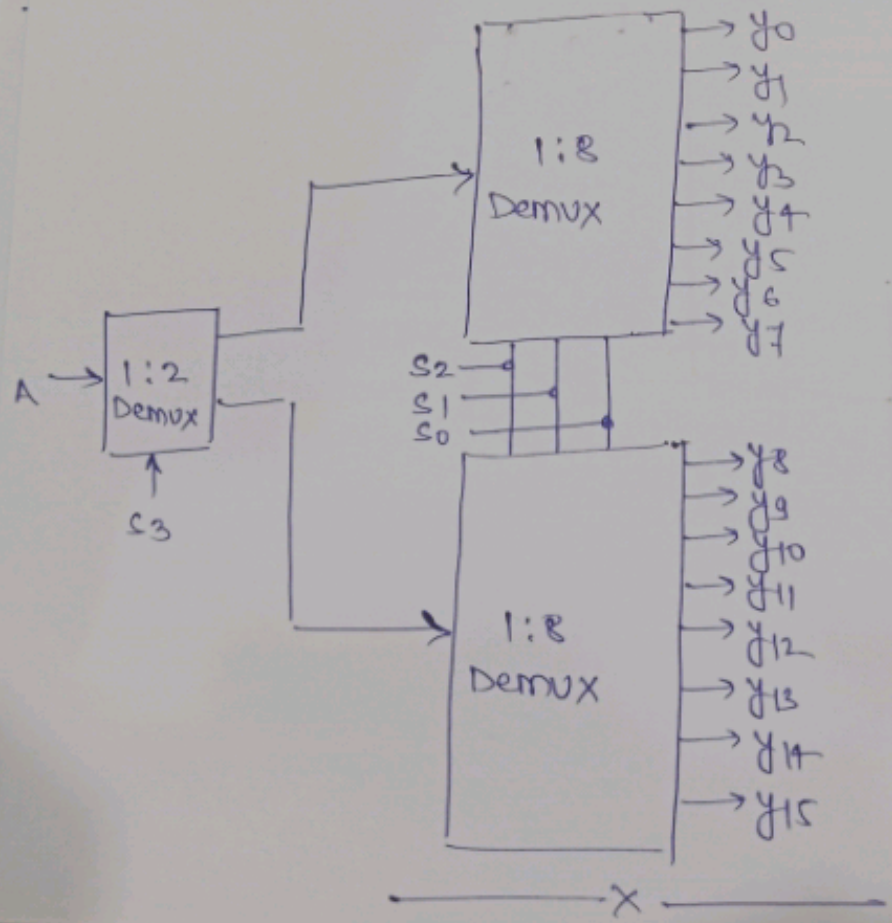
* Implement 8:1 MUX using 4:1 & 2:1 MUX :→ ②



I/P			O/P
S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

VIVO Y35

* Implement 1:8 Demux using 1:4 Demux and 1:2 Demux :→



* Implement 1:16 Demux using 1:8 and 1:2 Demux :→